

Medical Insurance Cost Analysis & Prediction

Exploring the interaction between body mass index (BMI) and smoking status using ordinary least squares regression and held-out test data.

1,337

cleaned observations

0.7418full-data interaction R^2 **\$5,594**

held-out test RMSE

Research question

Does the association between BMI and medical charges differ for smokers and non-smokers, and does modeling that interaction improve predictions on a held-out test set?

Data & preparation

The publicly shared Kaggle Medical Insurance Cost Dataset contains 1,338 records and seven fields: age, sex, BMI, children, smoker, region, and charges. There were no missing values. One pair of entirely identical rows was found; one row was removed for this educational analysis, leaving 1,337 observations. Without a person identifier, true duplicate identity cannot be confirmed.

Initial exploration

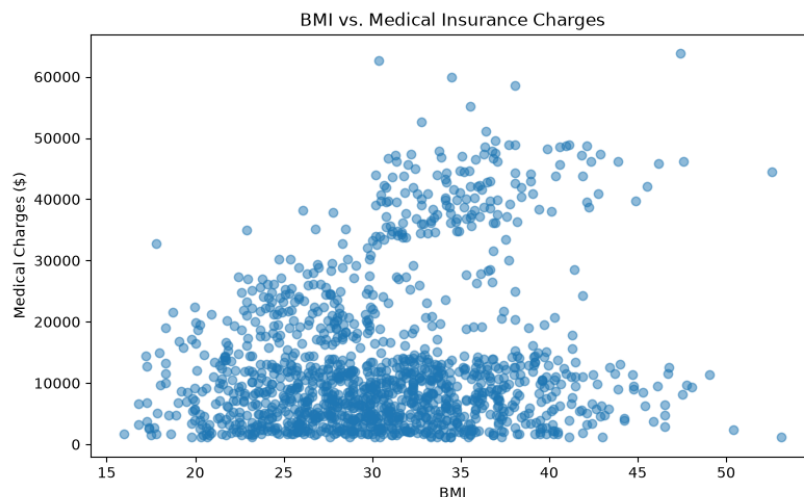


Figure 1. Original BMI-versus-charges scatterplot. Charges vary substantially at similar BMI values; this graph alone cannot establish causality.

Statistical modeling

Three nested ordinary least squares (OLS) specifications were first fitted to the full cleaned dataset for exploratory comparison.

Model structure

Model	Formula
1 - BMI only	charges ~ bmi
2 - Additive	charges ~ bmi + C(smoker)
3 - Interaction	charges ~ bmi * C(smoker)

Full-data model comparison

Model	R ²	Adjusted R ²	AIC
BMI only	0.0394	0.0386	28,884.01
BMI + Smoking	0.6579	0.6574	27,505.41
BMI × Smoking	0.7418	0.7412	27,131.23

Interaction coefficient estimates

Term	Coefficient (\$)	p-value
Intercept	5,886.14	< 0.001
Smoking (yes)	approx. -19,070*	< 0.001
BMI - non-smoker slope	83.34	0.008
BMI × Smoking	1,389.77	< 0.001

*The smoking coefficient was reported as -1.907e+04 in the console output, so its displayed decimal value is approximate. It refers to BMI = 0, outside the meaningful observed range.

The fitted BMI slope is approximately \$83.34 per BMI unit among non-smokers and \$1,473.11 among smokers (83.34 + 1,389.77). The interaction term indicates a difference in slopes in this sample, not a causal effect of smoking or BMI.

Source & reproducibility

Dataset: <https://www.kaggle.com/datasets/mosapabdelghany/medical-insurance-cost-dataset>

Analysis tools: Python, pandas, statsmodels, matplotlib, NumPy, and scikit-learn. Figures and metrics were supplied from the project run. This report does not independently rerun the original CSV or source code.

Interaction & prediction

The interaction model yields different fitted BMI slopes by smoking status. To assess predictive usefulness, the data were then split into new training and test partitions.

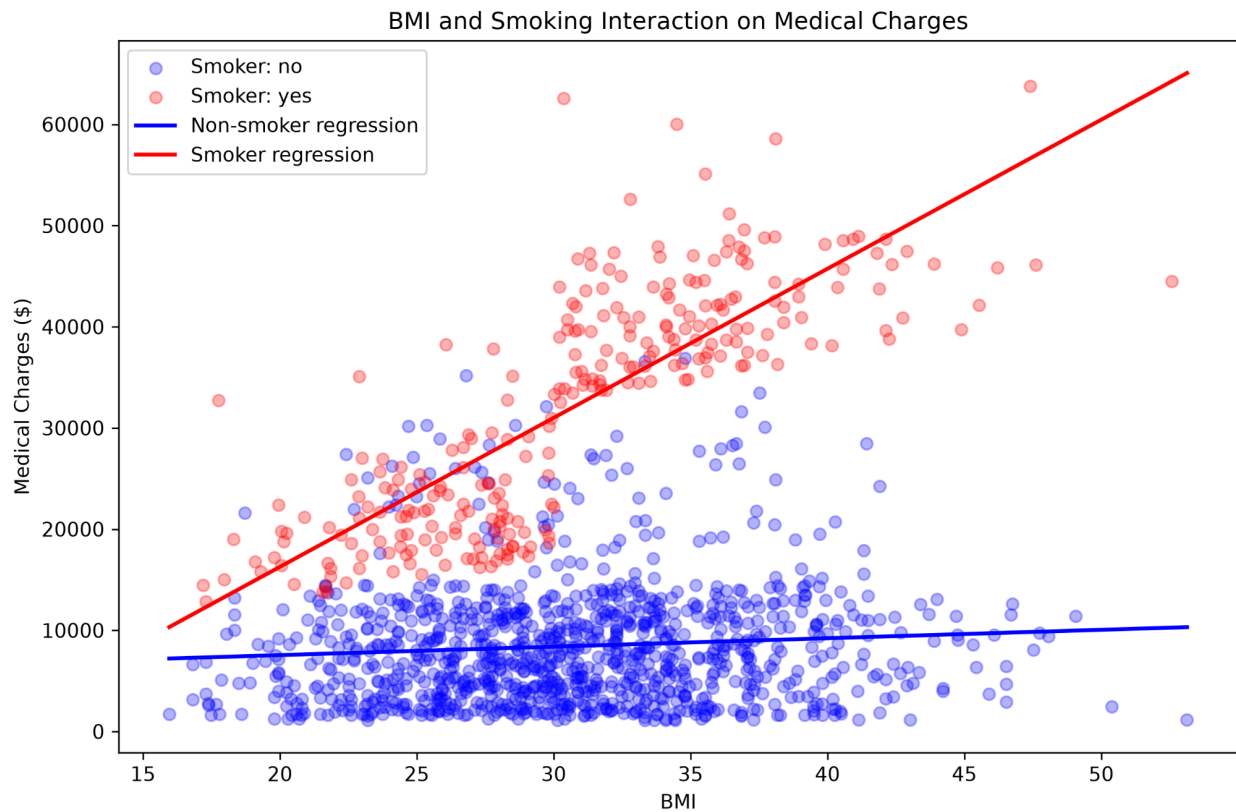


Figure 2. Observed charges by smoking status with group-specific fitted lines from the full-data exploratory interaction model.

Held-out evaluation

The cleaned records were divided into 1,069 training (80%) and 268 test (20%) observations via a random split stratified by smoking status (`random_state = 42`). Each specification was refitted on the training partition only. Metrics below are calculated on the test partition; they are not the full-data R^2 values shown on page 2.

Model	MAE (\$)	RMSE (\$)	Test R^2
BMI only	8,961.56	11,608.82	0.0651
BMI + Smoking	4,826.16	6,177.59	0.7353
BMI × Smoking	4,323.58	5,594.39	0.7829

In this single held-out split, the interaction model had lower MAE and RMSE than the additive specification. Relative to the additive model, test RMSE decreased by approximately 9.4%.

Interpretation & limitations

Prediction is informative at the aggregate level, but individual errors remain material.

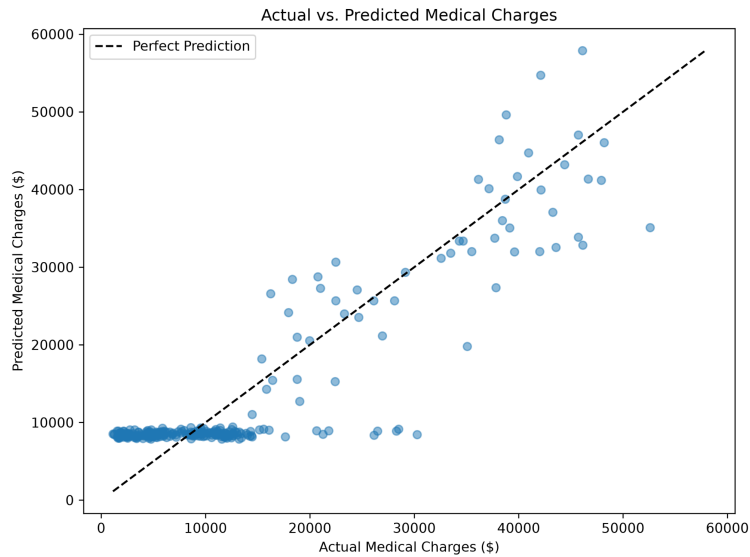


Figure 3. Held-out test observations: observed charges versus model predictions. The dashed diagonal represents exact predictions.

Key findings

BMI alone accounted for a small fraction of observed charges variation in the full sample ($R^2 = 0.0394$). Adding smoking status substantially improved fit ($R^2 = 0.6579$). Allowing a BMI-by-smoking interaction further improved the exploratory fit ($R^2 = 0.7418$) and reduced error on the held-out test observations (test RMSE = \$5,594.39; test $R^2 = 0.7829$).

Limitations & responsible interpretation

Data scope: These are charges, not an insurer-set premium or a verified future loss estimate; the public dataset lacks exposure, policy terms, claims development, and provenance sufficient for real pricing decisions. Its population may not represent current insureds.

Model scope: The two-predictor specification omits age, sex, region, children, and unrecorded health factors. Many test predictions cluster around \$8,000-\$9,000 while observed charges vary widely. OLS residual normality statistics also signal departure from normality; formal residual plots and heteroskedasticity checks were not completed. Standard coefficient p-values should be read with these assumptions in mind.

Validation scope: Only one 80/20 split was used. External data or repeated cross-validation would be needed to evaluate stability and generalizability. The associations are observational and cannot establish causation.

Possible next steps

Add a multivariable baseline, examine residuals, compare suitable positive-cost models (for example a Gamma GLM with log link), and validate with repeated splits or cross-validation before drawing practical pricing conclusions.